

EXAM II
NUMERICAL ANALYSIS
(MATH 461) FALL, 2023

DUE: 12/13/2023

Problem 1. Recall that the centered difference formula $D(h) = \frac{f(x+h) - f(x-h)}{2h}$ to approximate $f'(x)$ is derived from the fact that

$$f'(x) = D(h) + c_2 h^2 + c_4 h^4 + c_6 h^6 + \dots \quad (1)$$

- (a) Use Equation (1) to derive a formula, call it $D_2(h)$, approximating $f'(x)$ using the points $\pm h$ and $\pm 3h$ with a $\mathcal{O}(h^4)$ error term.
- (b) Let $D_2(h)$ be as above. Use Richardson extrapolation to derive a $\mathcal{O}(h^6)$ formula (expressed in terms of D_2) for computing $f'(x)$ by combining $D_2(h)$ and $D_2(3h)$.

Problem 2. Consider the generic initial value problem $y' = f(y)$, $y(t_0) = y_0$ (noting that, here, f is a function of y only) and the method

$$y_{n+1} = y_n + hf\left(y_n + \frac{h}{4}f(y_n)\right) + \tau_{n+1}$$

where τ_{n+1} is the local truncation error.

- (a) Find the leading order term of τ_{n+1} , i.e. an expression of the form $\tau_{n+1} = (\text{term}) + \mathcal{O}(h^{m+1})$.
- (b) State the order of the method we should expect using our rule-of-thumb heuristic (you do not need to provide proof of this).

Problem 3. (a) A simple approximation for $\int_0^h f(x)dx$ is $\int_0^h f(x)dx \approx hf(0)$. What is the error when expressed in terms of h and the derivative of f ?

- (b) Consider the approximation $\int_a^b f(x)dx \approx h \sum_{j=0}^{n-1} f(x_j)$ using the points $x_j = a + jh$ with $x_0 = a$ and $x_n = b$. Determine the error and express your answer in terms of n (not h).

HINT: By the mean value theorem, $f(x) = f(0) + f'(\zeta_x)x$ for some $\zeta_x \in (0, h)$. Use this and recall the mean value theorem for integrals is as follows: If f is continuous on $[a, b]$ and $g(x) \geq 0$ is an integrable function, then there exists $\xi \in (a, b)$ such that $\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$.

Problem 4. A central theme in this course has been finding approximation methods by solving linear algebra problems. In this problem, we consider yet another example of this. Consider the following scheme:

$$\beta f'_{i-2} + \alpha f'_{i-1} + f'_i + \alpha f'_{i+1} + \beta f'_{i+2} = c \frac{f_{i+3} - f_{i-3}}{6h} + b \frac{f_{i+2} - f_{i-2}}{4h} + a \frac{f_{i+1} - f_{i-1}}{2h},$$

where $f_i = f(x_i)$ and $f'_i = f'(x_i)$, and $h = x_{i+1} - x_i$ is the uniform grid size. The relation between the coefficients a, b, c and α, β are derived by matching the Taylor series coefficients of various orders. The first unmatched coefficients determines the formal truncation error of the scheme. Show that for the five unknowns we must have the following constraints:

- (a) $a + b + c = 1 + 2\alpha + 2\beta$ (second order)
- (b) $a + 2^2b + 3^2c = 2 \frac{3!}{2!}(\alpha + 2^2\beta)$ (fourth order)
- (c) $a + 2^4b + 3^4c = 2 \frac{5!}{4!}(\alpha + 2^4\beta)$ (sixth order)
- (d) $a + 2^6b + 3^6c = 2 \frac{7!}{6!}(\alpha + 2^6\beta)$ (eighth order)
- (e) $a + 2^8b + 3^8c = 2 \frac{9!}{8!}(\alpha + 2^8\beta)$ (tenth order),

Solving this (which you do not need to do) gives a tenth-order accurate approximation for the derivatives at each stencil point.